Question 1. 1. Proposition 3.1b in Humphreys states if I is a solveable ideal of L such that L/I is solveable, then L is solveable. Since $L/[L,L]$ is abelian by definition of $[L,L]$ and $[L,L]$ is nilpotent hence solveable we conclude by the proposition that L is solveable. □

2. We first note that $\mathrm{ad} : L \to \mathfrak{gl}(L)$ is a homomorphism, to see this we employ the Jacobi identity

$$[\mathrm{ad}_x, \mathrm{ad}_y](z) = \mathrm{ad}_x \mathrm{ad}_y(z) - \mathrm{ad}_y \mathrm{ad}_x(z) = [x,[y,z]] - [y,[x,z]] \overset{\text{Jacobi}}{=} [[x,y],z] = \mathrm{ad}_{[x,y]} \tag{1}$$

Thus by Humphreys Proposition 3.1(a) L being solveable implies $\mathrm{ad}(L)$ is a solveable subalgebra of $\mathfrak{gl}(L)$. Thus by Lie's theorem there exists a basis of L such that $\mathrm{ad}(L) \subset \mathfrak{t}(F, \dim L)$ under the identification of $\mathfrak{gl}(L)$ with $\mathfrak{gl}(F, \dim L)$. In part 4 of this problem we show that $[\mathfrak{t},\mathfrak{t}] = \mathfrak{n}$ the subalgebra of strictly upper triangular matrices. Thus it follows that

$$\mathrm{ad}[L,L] = [\mathrm{ad}\, L, \mathrm{ad}\, L] \subset [\mathfrak{t},\mathfrak{t}] = \mathfrak{n}(F, \dim L) \tag{2}$$

Since elements of $\mathfrak{n}(F, \dim L)$ are nilpotent, equation (2) tells us that $[L,L]$ is ad-nilpotent, thus by Engel's theorem $[L,L]$ is nilpotent. □

3. Consider the nonabelian 2 dimensional Lie algebra $L = \mathrm{span}_F(x,y)$ with relation $[x,y] = x$.

$$L^{(1)} = \mathrm{span}_F \{x\}, \quad L^{(2)} \overset{\text{def}}{=} [\mathrm{span}_F \{x\}, \mathrm{span}_F \{x\}] = 0 \tag{3}$$

However the central series has $L^1 = \mathrm{span}_F \{x\}$. It follows by induction that $L^k = [L, L^{k-1}] = [L, \mathrm{span}_F \{x\}] = \mathrm{span}_F(x)$ for all $k > 1$, hence L is not nilpotent. If you don't believe me that L is a lie algebra I give a calculation of the Jacobi identity.

$$[x,[x,y]] + [x,[y,x]] + [y,[x,x]] = [x,x] + [x,-x] + [y,0] = 0 + -[x,x] + 0 = 0 \tag{4}$$
$$[y,[x,y]] + [x,[y,y]] + [y,[y,x]] = [y,x] + [x,0] + [y,-x] = [y,x] + 0 - [y,x] = 0 \tag{5}$$

□

4. Throughout this exercise I will denote $\mathfrak{n}$ as the subset of strictly upper triangular matrices, and for $k \geq 1$ I will denote $\mathfrak{n}_k$ as the subset of matrices $\{\sum a_{ij} e_{ij} \mid a_{ij} \in F, j \geq i+k\}$. I will first compute the lower central series:

Claim 1. $[L,L] \subset \mathfrak{n}$

Claim 2. $[L,\mathfrak{n}] \supset \mathfrak{n}$

Claim 2 tells us in particular that $[L,L] \supset [L,\mathfrak{n}] \supset \mathfrak{n}$, so that using claim 1, $[L,L] = \mathfrak{n}$, it follows by induction and claim 1 that that $L^k = [L, L^{k-1}] = [L,\mathfrak{n}] = \mathfrak{n}$ is the lower central series.

The derived series requires sharper claims:

Claim 3. $[\mathfrak{n}_k, \mathfrak{n}_r] \subset \mathfrak{n}_{k+r}$

Claim 4. $[\mathfrak{n}_k, \mathfrak{n}_r] \supset \mathfrak{n}_{k+r}$

Using claims 3 and 4 and the solution for the lower central series we get the following derived series (note that for $k \geq n$ we have $\mathfrak{n}_k = 0$) $L^0 = \mathfrak{t}$, $L^1 = L^1 = \mathfrak{n}$ then by induction and using claims 3 and 4, we find for $k > 1$ that $L^{(k)} = [\mathfrak{n}_{2^{k-2}}, \mathfrak{n}_{2^{k-2}}] = \mathfrak{n}_{2^{k-1}}$. □

Proof of Claim 1. Let $A = \sum a_{ij} e_{ij}, B = \sum b_{ij} e_{ij} \in L$ and let $C = [A,B] = \sum c_{ij} e_{ij}$, then it suffices to show that $c_{jj} = 0$ for each j, this is straightforward since

$$c_{jj} = \sum_k a_{jk} b_{kj} - b_{jk} a_{kj} \tag{6}$$

but then since $a_{jk} = 0 = b_{jk}$ for $k < j$ and $b_{kj} = a_{kj} = 0$ for $k > j$ we find that

$$c_{jj} = a_{jj} b_{jj} - b_{jj} a_{jj} = 0 \tag{7}$$

□

Proof of Claim 2. It suffices to show that $e_{ij} \in [L, \mathfrak{n}]$ for all $j > i$, so let $j > i$ then $e_{ii} \in L$ and $e_{ij} \in \mathfrak{n}$ so that

$$[e_i i, e_{ij}] = e_{ii}e_{ij} - e_{ij}e_{ii} = e_{ij} \in [L, \mathfrak{n}] \tag{8}$$

□

Proof of Claim 3. Let $A = \sum a_{ij}e_{ij} \in \mathfrak{n}_s$ and $B = \sum a_{ij}e_{ij} \in \mathfrak{n}_r$, it suffices to show that $AB \in \mathfrak{n}_{s+r}$, because then by symmetry te same result will hold for BA and thus also their difference. So let $AB = \sum c_{ij}e_{ij}$, it follows that

$$c_{ij} = \sum_k a_{ik}b_{kj} \overset{A\in\mathfrak{n}_s}{=} \sum_{k\geq i+s} a_{ik}b_{kj} \overset{B\in\mathfrak{n}_r}{=} \sum_{j-r\geq k\geq i+s} a_{ik}b_{kj} \tag{9}$$

Thus the sum is only nonzero when $j \geq i+s+r$ thus $c_{ij} = 0$ for all $j < i+s+r$, this proves that $AB \in \mathfrak{n}_{s+r}$. □

Proof of Claim 4. It suffices to show that $e_{ij} \in [\mathfrak{n}_s, \mathfrak{n}_r]$ whenever $n \geq j \geq i+s+r$. Choose such an e_{ij}, then $e_{i,j-r} \in \mathfrak{n}_s$ and $e_{j-r,j} \in \mathfrak{n}_r$ so that

$$[e_{i,j-r}, e_{j-r,j}] = e_{i,j-r}e_{j-r,j} - \underbrace{e_{j-r,j}e_{i,j-r}}_{=0 \text{ since } j>i} = e_{ij} \in [\mathfrak{n}_s, \mathfrak{n}_r] \tag{10}$$

□

Question 2. 1. We first compute the matrices of ad_x and ad_y in $\mathfrak{gl}(L)$ with respect to the (ordered) basis $\{x, y\}$, this gives us

$$\mathrm{ad}_x = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \text{ and } \mathrm{ad}_y = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} \tag{11}$$

It follows that $\mathrm{ad}_x^2 = \mathrm{ad}_x$, $\mathrm{ad}_y^2 = 0$, $\mathrm{ad}_x \mathrm{ad}_y = \mathrm{ad}_y$ and $\mathrm{ad}_y \mathrm{ad}_x = 0$, by taking the trace of these matrices we directly read off the killing form: $\kappa(x, x) = 1$, $\kappa(y, y) = 0$ and $\kappa(x, y) = 0 = \kappa(y, x)$ thus relative to this basis the killing form ahs the following matrix

$$\kappa = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \tag{12}$$

We recall that $\mathrm{Rad}(\kappa) = \{z \in L \mid \kappa(z, -) \equiv 0\}$. Our prior computation shows that $\mathrm{Rad}(\kappa) = \mathrm{span}_F \{y\} = [L, L]$. □

2. Since the identity matrix $1_{n\times n} \notin \mathfrak{sl}(n)$ and $\dim \mathfrak{sl}(n) = n^2 - 1$ (see Humphreys section 1.1), we know that $\mathrm{span}_F \{1_{n\times n}\} \oplus \mathfrak{sl}(n)$ is a subspace of $\mathfrak{gl}(n)$ having dimension $n^2 - 1 + 1 = n^2 = \dim \mathfrak{gl}(n)$, hence is the entire vectorspace. Moreover if $x, y \in \mathfrak{gl}(n)$ symmetry of trace gives us that

$$\mathrm{tr}([x, y]) = \mathrm{tr}(xy) - \mathrm{tr}(yx) = \mathrm{tr}(xy) - \mathrm{tr}(xy) = 0 \tag{13}$$

in particular this holds for $x \in \mathfrak{sl}(n)$ so that $\mathfrak{sl}(n)$ is an ideal. Thus by part 4 of this problem, for $x, y \in \mathfrak{sl}(n)$ we have $\kappa_{\mathfrak{sl}(n)}(x, y) = \kappa_{\mathfrak{gl}(n)}(x, y)$ thus it suffices to compute the latter with respect to the basis $\{e_{ij}\}$ for $\mathfrak{gl}(V)$. Moreover, since κ is bilinear and trace is linear, it will suffice to show the identity holds on the basis $\{e_{ij} \mid i \neq j\} \cup \{e_{ii} - e_{i+1,i+1}\}$ of $\mathfrak{sl}(n)$. Symmetry of κ and trace imply that the following 3 cases are sufficient

Case 1. $x = e_{ij}, y = e_{kl}$*:* We first compute the action of $\mathrm{ad}_x \circ \mathrm{ad}_y$ on each $e_{st} \in \mathfrak{gl}(n)$.

$$[e_{ij}, [e_{k,\ell}, e_{st}]] = [e_{ij}, \delta_{\ell s}e_{kt} - \delta_{tk}e_{s\ell}] = \delta_{jk}\delta_{\ell s}e_{it} - \delta_{js}\delta_{tk}e_{i\ell} + \delta_{\ell i}\delta_{tk}e_{sj} - \delta_{ti}\delta_{\ell s}e_{kj} \tag{14}$$

When we take the trace of $\operatorname{ad}_x \operatorname{ad}_y$ we will sum over s,t the coefficients of the diagonal entries, so that the terms e_{ab} in the above sum are replaced with $\delta_{as}\delta_{bt}$, writing this explicitly we get

$$\begin{aligned}
\operatorname{tr}(\operatorname{ad}_x \operatorname{ad}_y) &= \sum_{s,t} \delta_{jk}\delta_{\ell s}\delta_{is} - \delta_{js}\delta_{tk}\delta_{is}\delta_{\ell t} + \delta_{\ell i}\delta_{tk}\delta_{jt} - \delta_{ti}\delta_{\ell s}\delta_{ks}\delta_{jt} && (15)\\
&= \sum_{s,t} \delta_{jk}\delta_{\ell s}\delta_{is} - \delta_{ij}\delta_{tk}\delta_{is}\delta_{\ell t} + \delta_{\ell i}\delta_{tk}\delta_{jt} - \delta_{ti}\delta_{\ell s}\delta_{k\ell}\delta_{jt} && (16)\\
&= \sum_{s,t} \delta_{jk}\delta_{\ell s}\delta_{is} - \underbrace{\delta_{ij}\delta_{tk}\delta_{is}\delta_{\ell t}}_{=0 \text{ since } i\neq j} + \delta_{\ell i}\delta_{tk}\delta_{jt} - \underbrace{\delta_{ti}\delta_{\ell s}\delta_{k\ell}\delta_{jt}}_{=0 \text{ since } k\neq \ell} && (17)\\
&= \sum_{s,t} \delta_{jk}\delta_{i\ell}\delta_{\ell s} + \sum_{s,t} \delta_{jk}\delta_{i\ell}\delta_{jt} && (18)\\
&= \sum_{\substack{1\leq t\leq n\\ s=\ell}} \delta_{jk}\delta_{i\ell} + \sum_{\substack{1\leq s\leq n\\ t=j}} \delta_{jk}\delta_{i\ell} && (19)\\
&= 2n\delta_{jk}\delta_{i\ell} && (20)\\
&= 2n \operatorname{tr}(e_{ij}e_{k\ell}) && (21)\\
&= 2n \operatorname{tr}(xy) && (22)
\end{aligned}$$

Case 2. $x = e_{ii} - e_{i+1,i+1}, y = e_{k\ell}$*:* Similarly to the above we first compute the action of $\operatorname{ad}_x \circ \operatorname{ad}_y$ on each e_{st}.

$$\begin{aligned}
[e_{ii} - e_{i+1,i+1}, [e_{k\ell}, e_{s,t}]] &= [e_{ii} - e_{i+1,i+1}, \delta_{\ell s}e_{kt} - \delta_{tk}e_{s\ell}] && (23)\\
&= \delta_{\ell s}\delta_{ik}e_{it} - \delta_{is}\delta_{tk}e_{i\ell} - \delta_{i+1,k}\delta_{\ell s}e_{i+1,t} + \delta_{i+1,s}\delta_{tk}e_{i+1,\ell} && (24)\\
&\quad - \delta_{ti}\delta_{\ell s}e_{ki} + \delta_{i\ell}\delta_{tk}e_{si} + \delta_{t,i+1}\delta_{\ell s}e_{k,i+1} - \delta_{\ell,i+1}\delta_{tk}e_{s,i+1} && (25)
\end{aligned}$$

We use the same reasoning as case 1 to compute the trace being

$$\begin{aligned}
\operatorname{tr}(\operatorname{ad}_x \operatorname{ad}_y) =& \sum_{s,t} \delta_{\ell s}\delta_{ik}\delta_{is} - \delta_{is}\delta_{tk}\delta_{is}\delta_{\ell t} - \delta_{i+1,k}\delta_{\ell s}\delta_{i+1,s} + \delta_{i+1,s}\delta_{tk}\delta_{i+1,s}\delta_{\ell t} && (26)\\
&+ \sum_{s,t} -\delta_{ti}\delta_{\ell s}\delta_{ks}\delta_{it} + \delta_{i\ell}\delta_{tk}\delta_{it} + \delta_{t,i+1}\delta_{\ell s}\delta_{k,s}\delta_{i+1,t} - \delta_{\ell,i+1}\delta_{tk}\delta_{i+1,t} && (27)\\
=& \sum_{s,t} \underbrace{\delta_{\ell s}\delta_{ik}\delta_{is}}_{0 \text{ since } \ell\neq k} - \underbrace{\delta_{tk}\delta_{is}\delta_{\ell t}}_{0 \text{ since } \ell\neq k} - \underbrace{\delta_{i+1,k}\delta_{\ell s}\delta_{i+1,s}}_{0 \text{ since } \ell\neq k} + \underbrace{\delta_{i+1,s}\delta_{tk}\delta_{\ell,t}}_{0 \text{ since } k\neq \ell} && (28)\\
&+ \sum_{s,t} - \underbrace{\delta_{\ell s}\delta_{ks}\delta_{it}}_{0 \text{ since } \ell\neq k} + \underbrace{\delta_{i\ell}\delta_{tk}\delta_{it}}_{0 \text{ since } \ell\neq k} + \underbrace{\delta_{\ell s}\delta_{ks}\delta_{i+1,t}}_{0 \text{ since } \ell\neq k} - \underbrace{\delta_{\ell,i+1}\delta_{tk}\delta_{i+1,t}}_{0 \text{ since } \ell\neq k} && (29)\\
=& 0 && (30)\\
=& 2n(\underbrace{\delta_{i\ell}\delta_{ik}}_{0 \text{ since } \ell\neq k} - \underbrace{\delta_{i+1,\ell}\delta_{i+1,k}}_{0 \text{ since } \ell\neq k}) && (31)\\
=& 2n \operatorname{tr}((e_{ii} - e_{i+1,i+1})e_{k\ell}) && (32)\\
=& 2n \operatorname{tr}(xy) && (33)
\end{aligned}$$

Case 3. $x = e_{ii} - e_{i+1,i+1}, y = e_{jj} - e_{j+1,j+1}$*:* TODO

3. By theorem 5.1, it suffices to show that the killing form is non-degenerate, i.e. every $x \in \mathfrak{sl}(n)$ has some $y \in \mathfrak{sl}(n)$ such that $\kappa(x,y) = 2n\operatorname{tr}(xy) \neq 0$, write $x = \sum x_{ij}e_{ij}$, then if $x_{ij} \neq 0$ for some $i \neq j$, then

$$xe_{ji} = \sum_{k,\ell} x_{k,\ell}e_{k\ell}e_{ji} = \sum_{k} x_{kj}e_{ki} \tag{34}$$

So that $\operatorname{tr}(xe_{ji}) = x_{ij} \neq 0$. Otherwise we can assume x is diagonal, then since $x \neq 0$ there is some diagonal entry $x_{ii} \neq 0$, furthermore since $\operatorname{tr}(x) = 0$ there must be some other diagonal entry x_{jj} with $x_{jj} \neq x_{ii}$. It follows that

$$x(e_{ii} - e_{jj}) = x_{ii}e_{ii} - x_{jj}e_{jj} \tag{35}$$

and $\operatorname{tr}(x(e_{ii} - e_{jj})) = x_{ii} - x_{jj} \neq 0$. Thus in either case we can choose $y = e_{ji}$ or $y = (e_{ii} - e_{jj})$ so that $\operatorname{tr}(xy) \neq 0$. Now applying the previous subpart we find that

$$\kappa(x, y) = 2n \operatorname{tr}(xy) \neq 0 \tag{36}$$

We conclude that the killing form is nondegenerate and thus $\mathfrak{sl}(n)$ is semisimple by Humphreys Theorem 5.1. □

4. Since $I \subset L$ is a subspace, it has a compliment S satisfying $L = I \oplus S$ as a vector space. We may take $v_1, \ldots, v_r$ to be a basis for I and $v_{r+1}, \ldots, v_n$ to be a basis for S. Whence any element of $\mathfrak{gl}(L)$ can be written in the form $\sum a_{ij}e_{ij}$ for $a_{ij} \in F$ and $e_{ij}(v_k) = \delta_{jk}v_i$. Now let $x, y \in I$, since I is an ideal we know that $\operatorname{ad}_x, \operatorname{ad}_y : L \to I$, so in particular so does their compositon. In terms of our basis this tells us that for $\operatorname{ad}_x = \sum x_{ij}e_{ij}, \operatorname{ad}_y = \sum y_{ij}e_{ij}$ and $\operatorname{ad}_x \circ \operatorname{ad}_y = \sum a_{ij}e_{ij}$ we have $x_{ij} = y_{ij} = a_{ij} = 0$ for $i > r$, whence in particular $e_{ii} = 0$ for $i = r+1, \ldots, n$ it follows that

$$\kappa(x, y) = \operatorname{tr}(\operatorname{ad}_x \circ \operatorname{ad}_y) = \sum_1^n a_{ii} = \sum_1^r a_{ii} \tag{37}$$

$$= \sum_{i=1}^r \sum_{k=1}^n x_{ik}y_{ki} = \sum_{i=1}^r \sum_{k=1}^r x_{ik}y_{ki} = \operatorname{tr}(\operatorname{ad}_x|_I \circ \operatorname{ad}_y|_I) = \kappa_I(x, y) \tag{38}$$

So for any $(x, y) \in I \times I$ we have $\kappa(x, y) = \kappa_I(x, y)$, this of course implies that $\kappa|_{I\times I} = \kappa_I$. □

Now to see that this result does not hold for subalgebras consider the nonabelian subalgebra of dimension 2 given in teh first part of the question, and consider the subalgebra $H = \operatorname{span}_F\{x\}$, we computed in part 1 that $\kappa(x, x) = 1$ whence $\kappa|_{H\times H}$ has matrix 1, but since H is abelian ad_x acts as zero on H so that $\kappa_H(x, x) = 0$, so that κ_H has matrix 0. □

Question 3. 1. Since toral subalgebras are abelian (Humphreys Lemma 8.1) it suffices to show that H is maximal abelian. Using the computation from Humphreys section 1.2 we know that matrices in $\mathfrak{sp}(4)$ are of the following form

$$\begin{pmatrix} A & B \\ C & -A^t \end{pmatrix} \tag{39}$$

where $A, B, C \in \mathfrak{gl}(2)$ and B, C are symmetric. Now we take $x = \operatorname{Diag}(1, 1, -1, -1) \in H$ and compute

$$\left[x, \begin{pmatrix} A & B \\ C & -A^t \end{pmatrix}\right] = \begin{pmatrix} 0 & 2B \\ -2C & 0 \end{pmatrix} \tag{40}$$

so that an abelian subalgebra containing H must only contain matrices of the form $(A, 0; 0, -A^t)$. We will be done once we show that A need be diagonal for such an element to be included in H. So choose a non-diagonal matrix $A = (a, b; c, d)$ for b, c not both zero, then $x = \operatorname{Diag}(1, -1, -1, 1) \in H$ and

$$\left[x, \begin{pmatrix} A & 0 \\ 0 & -A^t \end{pmatrix}\right] = \begin{pmatrix} \begin{pmatrix} 0 & 2b \\ -2c & 0 \end{pmatrix} & 0 \\ 0 & \begin{pmatrix} 0 & 2c \\ -2b & 0 \end{pmatrix} \end{pmatrix} \neq 0 \tag{41}$$

It follows that any abelian subalgebra containing H contains only diagonal matrices, but H already contains all the diagonal matrices in $\mathfrak{sp}(2)$ this shows that H is maximal abelian and thus maximal toral by Humphreys Lemma 8.1. □

2. We take the basis $h_1 = e_{11} - e_{33}$ and $h_2 = e_{22} - e_{44}$ for H. The roots space decomposition is as follows:

$$L_{\beta_1} = \operatorname{span}_F \{e_{12} - e_{43}\}, \beta_1(h_1) = 1, \beta_1(h_2) = -1 \tag{42}$$
$$L_{\beta_2} = \operatorname{span}_F \{e_{21} - e_{34}\}, \beta_2(h_1) = -1, \beta_2(h_2) = 1 \text{ so that } \beta_2 = -\beta_1 \tag{43}$$
$$L_{\beta_3} = \operatorname{span}_F \{e_{14} + e_{23}\}, \beta_3(h_1) = 1, \beta_3(h_2) = 1 \tag{44}$$
$$L_{\beta_4} = \operatorname{span}_F \{e_{32} + e_{41}\}, \beta_4(h_1) = -1, \beta_4(h_2) = -1 \text{ so that } \beta_4 = -\beta_3 \tag{45}$$
$$L_{\beta_5} = \operatorname{span}_F \{e_{13}\}, \beta_5(h_1) = 2, \beta_5(h_2) = 0 \tag{46}$$
$$L_{\beta_6} = \operatorname{span}_F \{e_{31}\}, \beta_6(h_1) = -2, \beta_6(h_2) = 0 \text{ so that } \beta_6 = -\beta_5 \tag{47}$$
$$L_{\beta_7} = \operatorname{span}_F \{e_{24}\}, \beta_7(h_1) = 0, \beta_7(h_2) = 2 \tag{48}$$
$$L_{\beta_8} = \operatorname{span}_F \{e_{42}\}, \beta_8(h_1) = 0, \beta_8(h_2) = -2 \text{ so that } \beta_8 = -\beta_7 \tag{49}$$

Writing explicitly the roots as linear functionals let $A = \operatorname{Diag}(a, b, -a, -b) \in H$, then

$$\beta_1(A) = a - b = -\beta_2(A) \tag{50}$$
$$\beta_3(A) = a + b = -\beta_4(A) \tag{51}$$
$$\beta_5(A) = 2a = -\beta_6(A) \tag{52}$$
$$\beta_7(A) = 2b = -\beta_8(A) \tag{53}$$

□

3. Take $\alpha_1 = \beta_1$ in the previous subpart and $\alpha_2 = \beta_7$, this lets us rewrite the root system as

$$\Phi = \{\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1 + \alpha_2), \pm(2\alpha_1 + \alpha_2)\} \tag{54}$$

To compute the Cartan matrix we use the computations of the elements t_α from the fourth subpart, as these are needed to take inner products.

4. Humphreys proposition 8.3 says that for $x \in L_\alpha$, $y \in L_{-\alpha}$ we have $[x, y] = \kappa(x, y)t_\alpha$. We need the following claims:

Claim 1. If $h, h' \in H$ then $\kappa(h, h') = 6\operatorname{tr}(hh')$

Claim 2. Let $\alpha \in \Phi$ then take root vectors $x_\alpha \in L_\alpha$, $y_\alpha \in L_{-\alpha}$ and $h \in \Phi$, then

$$\kappa(x_\alpha, y_\alpha) = \frac{6\operatorname{tr}(h[x_\alpha, y_\alpha])}{\alpha(h)} \tag{55}$$

Using the claims we find that for $\alpha \in \Phi$

$$t_\alpha = \frac{[x_\alpha, y_\alpha]}{\kappa(x_\alpha, y_\alpha)} = \frac{\alpha(h)[x_\alpha, y_\alpha]}{6\operatorname{tr}(h[x_\alpha, y_\alpha])} \tag{56}$$

Thus we compute each $[x_\alpha, y_\alpha]$, $h[x_\alpha, y_\alpha]$ and $\alpha(h)$ where $h \in \{h_1, h_2\}$ so that $\alpha(h) \neq 0$ below:

$$[x_{\alpha_1}, y_{\alpha_1}] = [e_{12} - e_{43}, e_{21} - e_{34}] = e_{11} + e_{44} - e_{22} - e_{33} = h_1 - h_2 \tag{57}$$
$$[x_{\alpha_2}, y_{\alpha_2}] = [e_{24}, e_{42}] = e_{22} - e_{44} = h_2 \tag{58}$$
$$[x_{\alpha_1+\alpha_2}, y_{\alpha_1+\alpha_2}] = [e_{14} + e_{23}, e_{32} + e_{41}] = e_{11} + e_{22} - e_{33} - e_{44} = h_1 + h_2 \tag{59}$$
$$[x_{2\alpha_1+\alpha_2}, y_{2\alpha_1+\alpha_2}] = [e_{13}, e_{31}] = e_{11} - e_{33} = h_1 \tag{60}$$
$$h_1[x_{\alpha_1}, y_{\alpha_1}] = e_{11} + e_{33} \tag{61}$$
$$h_2[x_{\alpha_2}, y_{\alpha_2}] = e_{22} + e_{44} \tag{62}$$
$$h_1[x_{\alpha_1+\alpha_2}, y_{\alpha_1+\alpha_2}] = e_{11} + e_{33} \tag{63}$$
$$h_1[x_{2\alpha_1+\alpha_2}, y_{2\alpha_1+\alpha_2}] = e_{11} + e_{33} \tag{64}$$
$$\alpha_1(h_1) = 1 \tag{65}$$
$$\alpha_2(h_2) = 2 \tag{66}$$
$$(\alpha_1 + \alpha_2)(h_1) = 1 \tag{67}$$
$$(2\alpha_1 + \alpha_2)(h_1) = 2 \tag{68}$$

Whence we use the formula from equation (56) to compute each t_α:

$$t_{\alpha_1} = \frac{\alpha_1(h_1)(h_1 - h_2)}{6\,\mathrm{tr}(e_{11} + e_{33})} = \frac{1}{12}(h_1 - h_2) \tag{69}$$
$$t_{\alpha_2} = \frac{\alpha_2(h_2)h_2}{6\,\mathrm{tr}(e_{22} + e_{44})} = \frac{1}{6}h_2 \tag{70}$$
$$t_{\alpha_1+\alpha_2} = \frac{(\alpha_1 + \alpha_2)(h_1)(h_1 + h_2)}{6\,\mathrm{tr}(e_{11} + e_{33})} = \frac{1}{12}(h_1 + h_2) \tag{71}$$
$$t_{2\alpha_1+\alpha_2} = \frac{(2\alpha_1 + \alpha_2)(h_1)h_1}{6\,\mathrm{tr}(e_{11} + e_{33})} = \frac{1}{6}h_1 \tag{72}$$

Now to compute the coroots, we use Claim 1 to compute

TODO: Prove the claims